



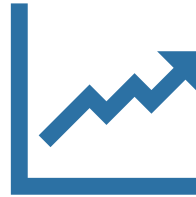
INTEGRATED ESG ANALYTICS: A DATA-DRIVEN FRAMEWORK FOR SUSTAINABLE INVESTMENT DECISIONS

- BY SOMSUVRA NATH

Objective



Develop a data-driven ESG analytics framework using SQL



To empower Investment Analysts to identify sustainable investment opportunities, benchmarks industry performance



To generate actionable buy/sell signals using SQL analysis of financial and sustainability metrics

Data & Methodology

Overview of Table

- **20 unique companies** across industries (Retail, Technology, Transportation, Healthcare, Energy, Finance, Utilities, Consumer Goods)
- **Regions:** Latin America, Asia, Africa, Middle East, Oceania, Europe, North America
- **Time span:** 2015–2025 (yearly data)
- **Key metrics:**
 - Financial: Revenue, Profit Margin, Market Cap, Growth Rate
 - ESG: Overall score + Environmental, Social, Governance sub-scores

Source: [esgfinancialdata.xlsx](#)

Tools Used: SQL, Excel

Analytical Framework

- Screening & Filtering
- Ranking & Benchmarking
- Trend Analysis
- Comparative Analysis

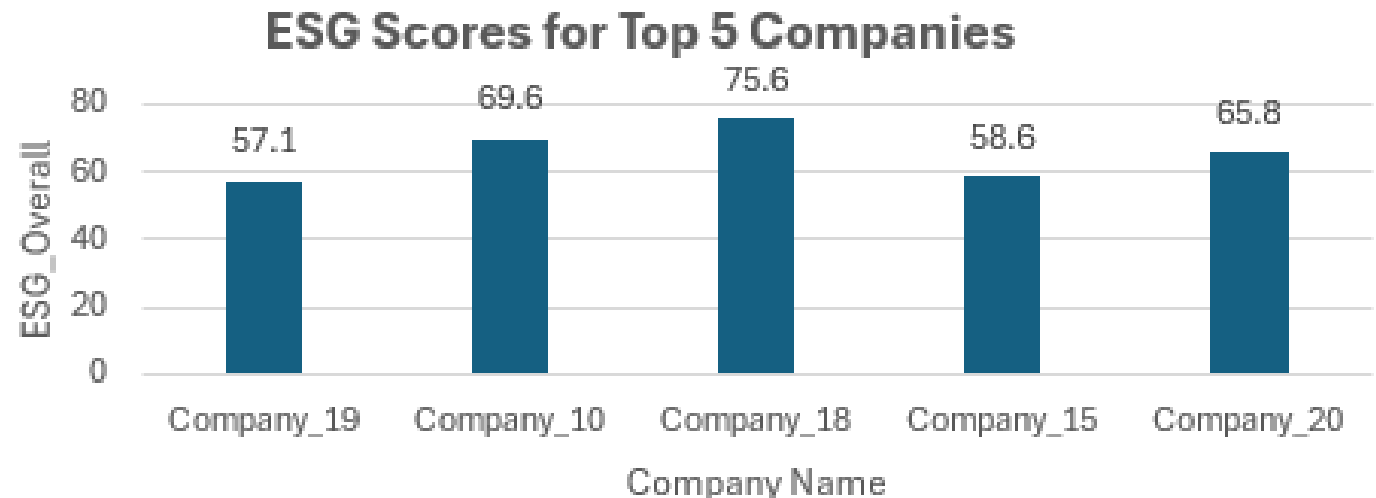
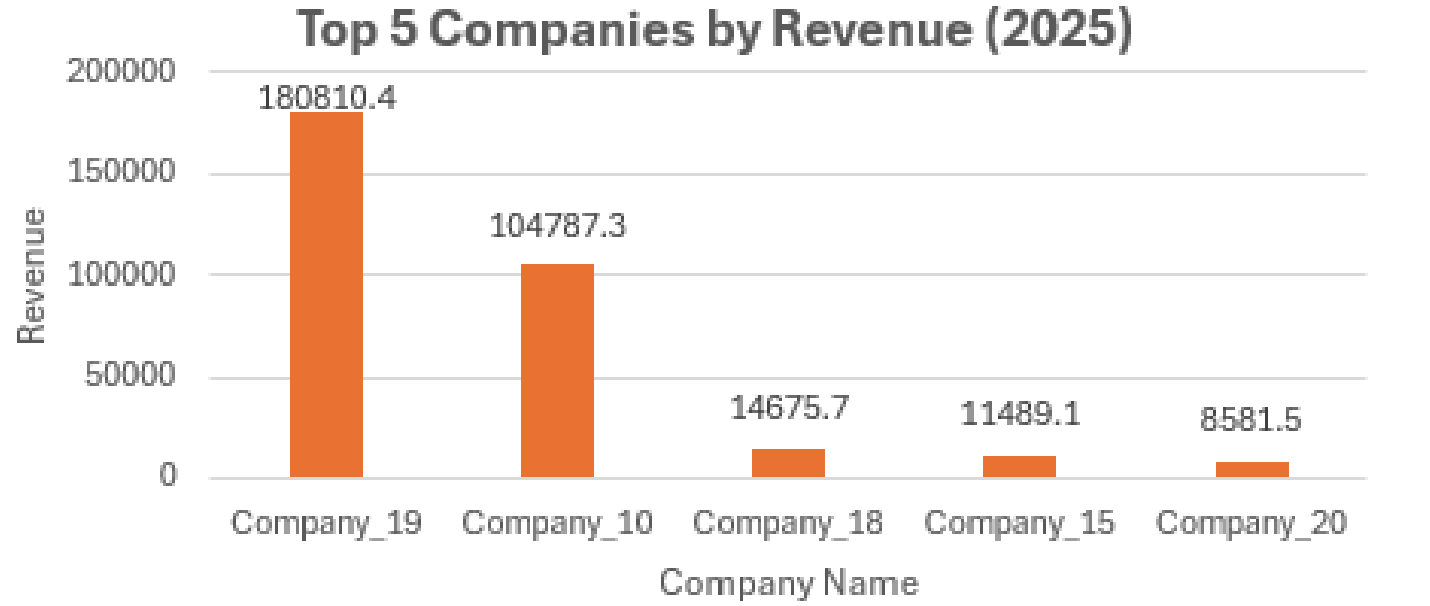
Key Metrics: Revenue, Market Cap, ESG Scores

SQL Code: https://github.com/somsuvranath/SQL_ESG

This analysis uses a simulated dataset. Company names, financials, and ESG scores are fictitious and created for the purpose of demonstrating data processing, ESG evaluation, and investment scenario modelling.

Do big players lead in ESG?

- The opposite trend appears here — the largest company by revenue has the **lowest ESG score**.
- Mid-sized Companies Show **Stronger ESG**
- Larger companies may face more complex ESG challenges across global operations.
- Smaller firms might be more agile or mission-driven, integrating ESG earlier.
- **Investors & Stakeholders** should look for companies like **Company_10**, which balance strong revenue with solid ESG performance.



Potential ESG Investment Opportunities

Most Attractive ESG Opportunity

Company_18

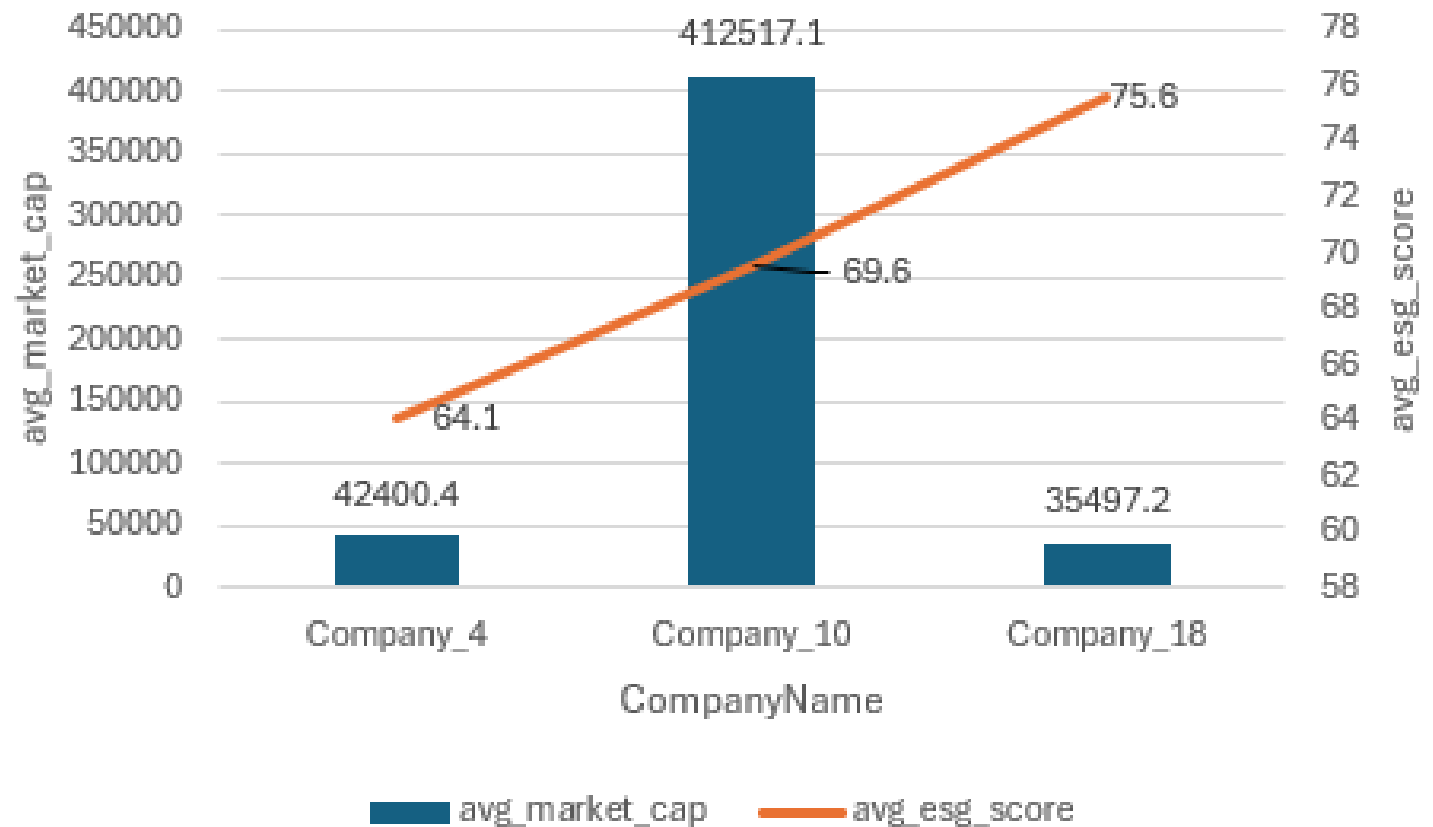
- **Highest ESG score (75.6)** → Leader in ESG performance
- **Smallest market cap** → Potential for growth, possibly undervalued
- **Strong ESG could indicate** good governance, lower risk, and future resilience

Balanced & Stable ESG Opportunity

Company_10

- **Strong ESG (69.6)** → Well above average
- **Largest market cap** → Established, stable, lower volatility
- **Suitable for** conservative ESG investors seeking size + sustainability

Above-average Market Cap and above-average ESG scores in 2025



ESG Leaders and Laggards in 2025

ESG Leaders

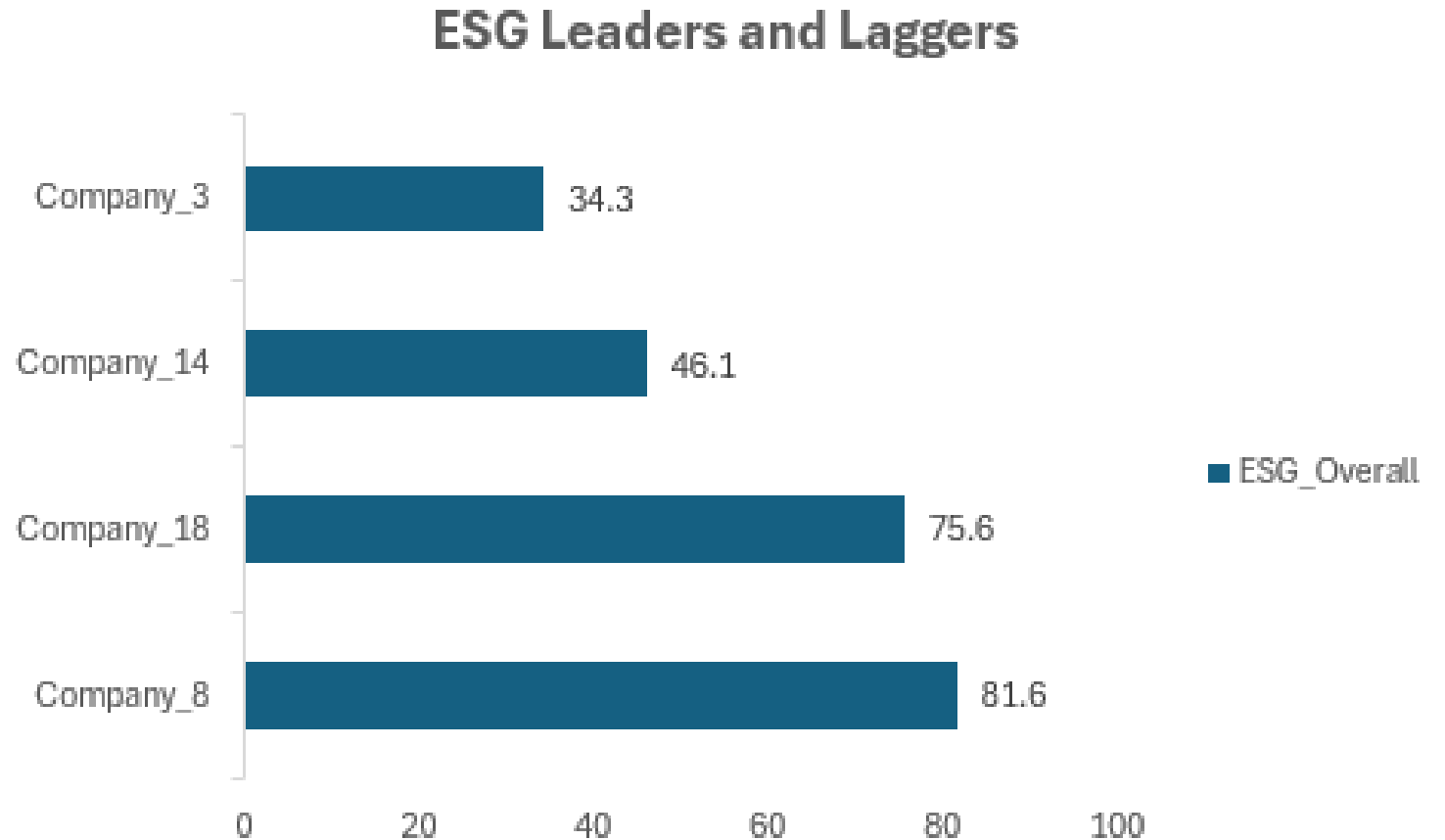
- Company_8 and Company_18

ESG Laggards

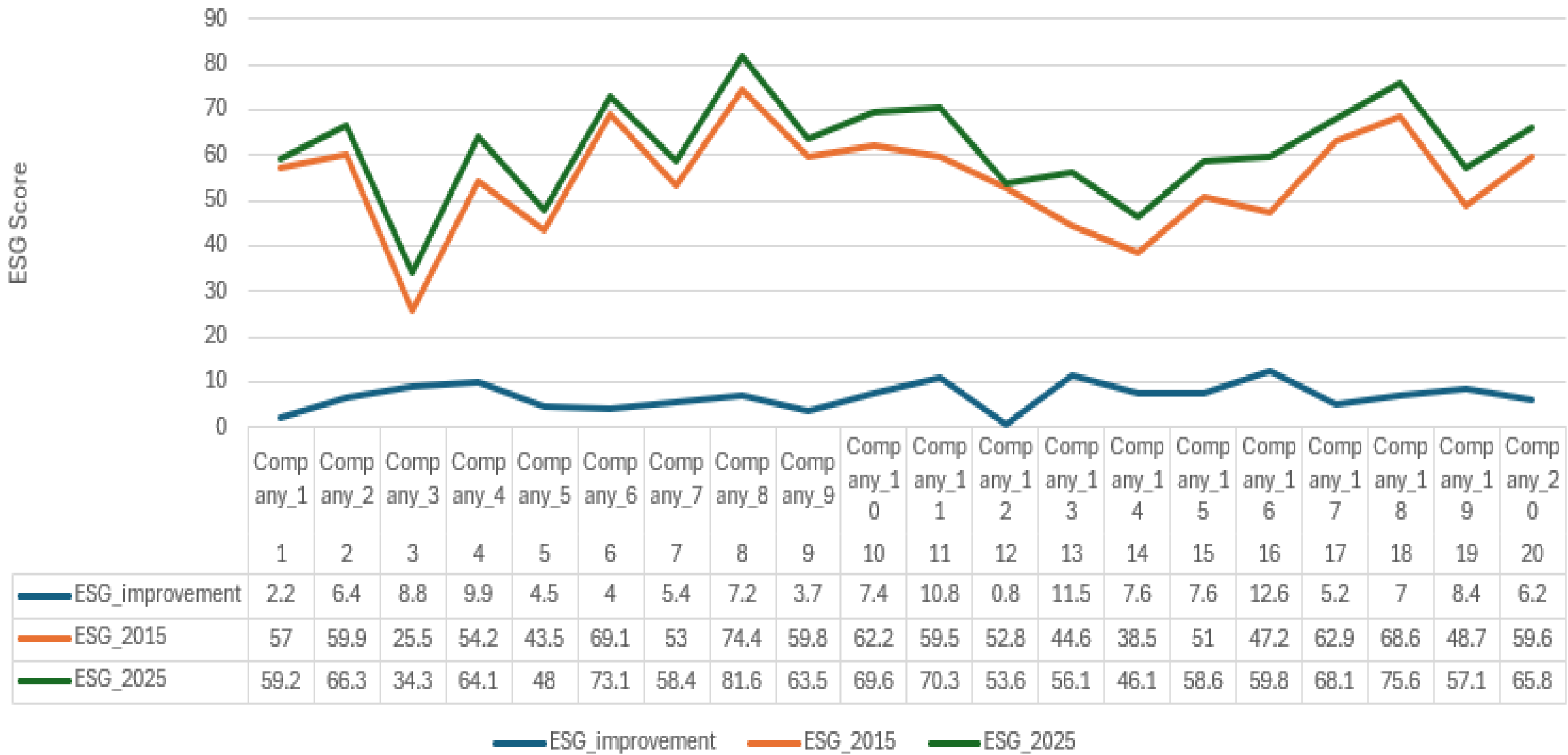
- Company_14 and Company_3

For Investors

- **Prioritize** Company_8 and Company_18 for ESG-focused portfolios
- **Exercise caution** with Company_3 unless ESG transformation is underway
- **Engage** with Company_14 on ESG improvement plans



ESG Improvement from 2015 to 2025



Improvement Analysis – Who's Making Progress?

Positive Findings

- **High Improvers Often Started Mid/Low : Company_16** (47.2 → 59.8) and **Company_13** (44.6 → 56.1) made the largest gains
- **ESG Leaders Maintain Momentum : Company_8** (74.4 → 81.6) stayed #1 while still improving
- **Strong Improvement with Solid Base : Company_11** jumped from 59.5 → 70.3 (+10.8)

Areas of Concern

- **Stagnation Risk: Company_12** barely moved (+0.8)
- **Low Base, Still Low: Company_3** improved +8.8 but remains **last** (34.3)
- **High Start, Low Growth: Company_6** started high (69.1) but improved only by +4.0

For Investors

- **Invest in momentum:** Companies like **16, 13, 11, 4** show a strong upward trajectory
- **Hold leaders:** **Company_8 & 18** deliver consistent high performance
- **Avoid stagnation:** **Company_12 & 1** show a lack of ESG commitment
- **Engage with laggards:** **Company_3** needs transformation; assess if turnaround is credible

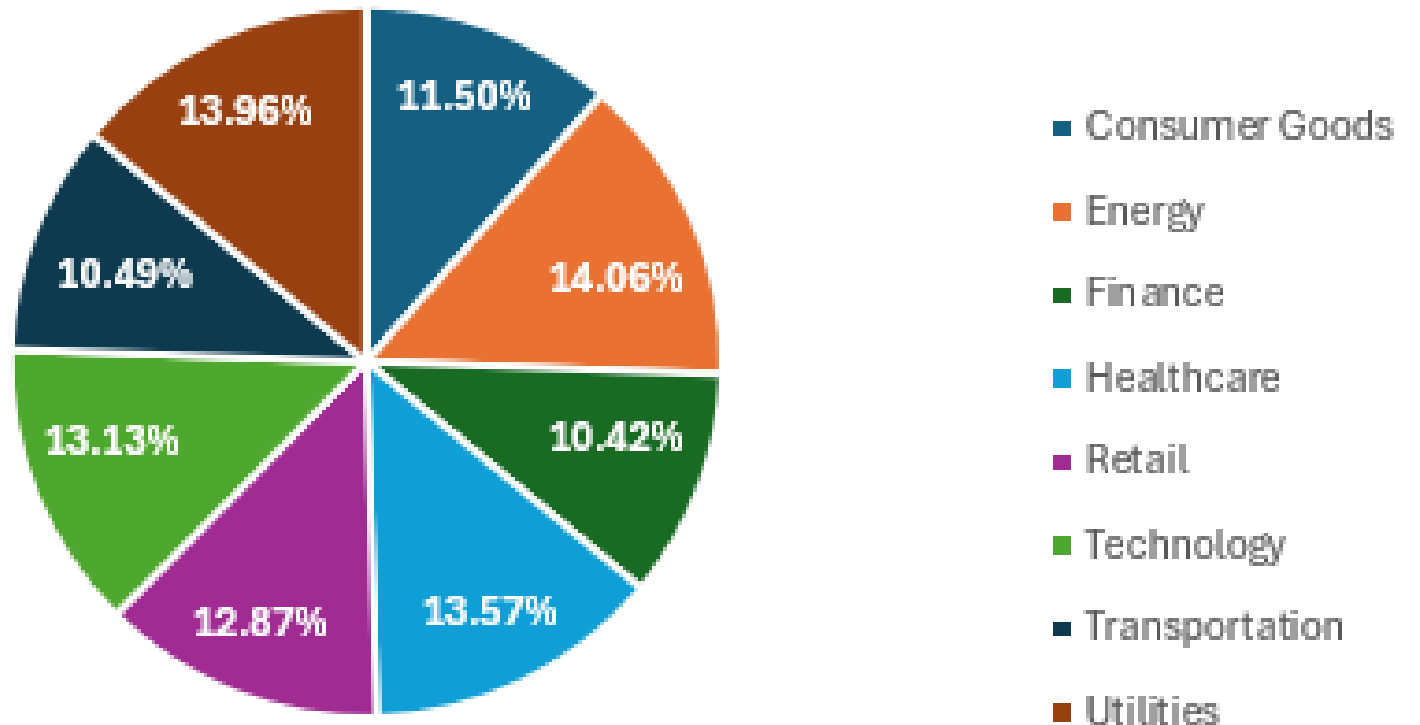
Industry Share in 2025

For Investors

- **Overweight Energy & Utilities** for stability and size
- **Consider Healthcare** for growth + stability blend
- **Technology** offers innovation, but isn't the largest slice
- **Finance** may offer niche opportunities despite a small share

The market is **led by essential services (Energy, Utilities, Healthcare)**, not by **high-growth tech** alone. This suggests a **mature, balanced economy** where infrastructure and basic needs drive the largest market segments.

Market Share Distribution by Industry in 2025



Regional Comparison - Regulatory Impact Analysis

For Investors

- For higher ESG returns → Look to **Europe** (higher ceiling, more variance)
- For ESG stability → **North America** (consistent, predictable scores)
- **Top performer globally** is in **Europe** (Company scoring 81.6)

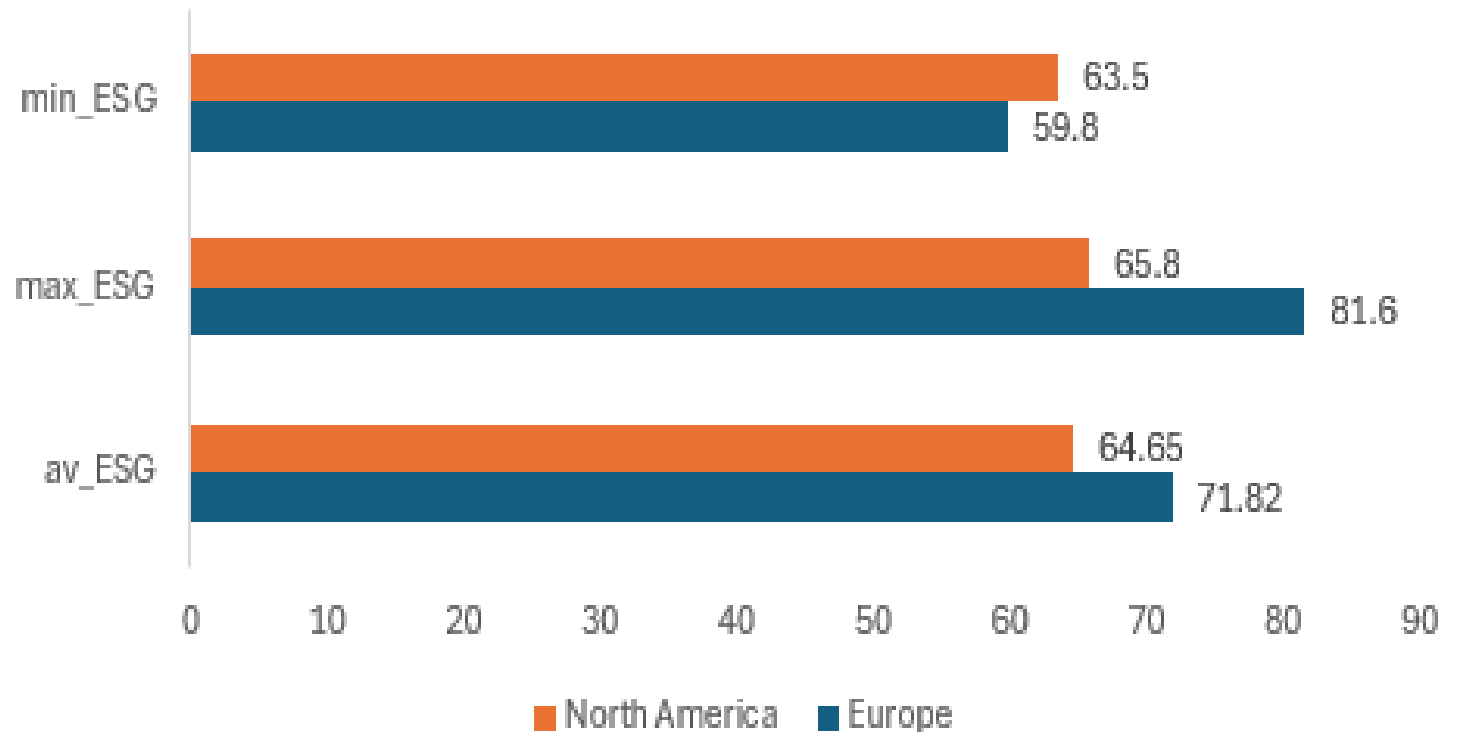
Europe

- Ahead in **ESG leadership** but has **greater internal disparity**.

North America

- More **uniform** but **lags in top-tier performance**.

ESG Score Comparison between North America and Europe in 2025



KEY TAKEAWAYS

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- **ESG as a Performance Marker:** Companies with **higher ESG scores do not consistently have higher revenues**. For example, **Company_18** has the highest ESG score (75.6) but only moderate revenue, while **Company_1** has the highest revenue but a middling ESG score (57.1). This suggests that strong current revenue is not dependent on top-tier ESG performance, though ESG may influence long-term stability.
- **Significant ESG Improvers:** **Company_18** is a clear standout, achieving both the **highest 2025 ESG score (75.6)** and the **largest ESG improvement (+78.4 points)** between 2015 and 2025. This indicates a successful, transformative corporate responsibility strategy. Conversely, **Company_17** showed a major decline (-5.2 points), signalling potential risk or governance issues.
- **Regional ESG Leadership:** The European market shows a clear ESG advantage. The **maximum (81.6)** and **average (71.82)** ESG scores in Europe are substantially higher than in North America (max 65.8, avg 64.65). This reflects stricter regulatory environments or more advanced sustainability practices in European companies.
- **Market Concentration:** The **Energy** and **Utilities** sectors hold the largest combined market share (**28.02%**), indicating these traditional, high-impact industries continue to dominate the market landscape in 2025.



CONCLUSION

- ESG performance has become a significant but complex factor in the corporate landscape, showing strong regional and sectoral trends, yet its direct financial payoff varies widely and does not guarantee market leadership.
- The data clearly shows a strong trend toward ESG adoption, particularly in Europe, but this is not yet a universal driver of financial success.



**Do you have any
Questions?**





THANK YOU

